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POLICY AT A GLANCE | 05

An insider look into Health and Disaster Risk Management Situation and Preparedness at Sub-national level

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Health Sector in India

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Policy at a glance

An insider look into Health and Disaster Risk Management Situation and Preparedness at Sub-national level in India

Health Sector in India

The '**Policy-at-a-Glance**' provides users with curated information for first-level analysis and inter-state comparisons of policy. The document brings together all topic-relevant national and state-level acts, policies, and schemes in one place. Further, it identifies for a reader what policy features are covered in which document in a simple table.

The first section of this document sets out some '*Context and Background*' on the topic; the subsequent sections explain the '*Policy Features*' that have been compared; and the last section is the '*Table of Insights*' that marks out what '*Policy Feature*' has been covered (and on which page) in each of the topic-relevant documents.

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1. Context

Health Emergency Preparedness (HEP) is the ability of a health system to prevent, detect, respond to, and recover from public health emergencies, such as epidemics, pandemics, natural disasters, and bioterrorism. HEP is essential for safeguarding the health and well-being of the population, as well as for ensuring the continuity of essential health services and functions.

India has faced several public health emergencies in the past, such as the 1994 plague outbreak in Surat, the 2001 Gujarat earthquake, the 2004 Indian Ocean tsunami, the 2005 Kashmir earthquake, the 2009 H1N1 influenza pandemic, the 2013 Uttarakhand floods, the 2014 Odisha cyclone, the 2015 Chennai floods, the 2016 Zika virus outbreak, and the recent 2020 COVID-19 pandemic. These emergencies have highlighted the areas of improvement in India's health system preparedness and response capacity, such as infrastructure, human resources, surveillance, laboratory network, logistics, coordination, communication, and financing.

To address these gaps and challenges, India has adopted various policies and programs at the national and state levels for HEP. Some of the key policies and programs are:

- **The National Health Policy (NHP) 2017:** The NHP 2017 provides a comprehensive framework for achieving universal health coverage and delivering quality healthcare services to all at affordable cost. The NHP 2017 recognizes HEP as one of the key policies thrust areas and envisages strengthening the public health system for effective management of health emergencies.
- **The National Disaster Management Plan (NDMP) 2019:** The NDMP 2019 provides a holistic and integrated approach to disaster risk reduction and management in India. The NDMP 2019 covers all phases of disaster management cycle- prevention, mitigation, preparedness, response, recovery, and reconstruction. The NDMP 2019 also outlines the roles and responsibilities of various stakeholders, including the health sector.
- **The Integrated Disease Surveillance Programme (IDSP):** The IDSP is a flagship program of the Ministry of Health and Family Welfare (MoHFW) that aims to strengthen disease surveillance and outbreak response in India. The IDSP collects and analyses data on various communicable and non-communicable diseases from different sources, such as laboratories, hospitals, media, and community. The IDSP also provides timely information and alerts to the health authorities for appropriate public health actions.
- **The National Centre for Disease Control (NCDC):** The NCDC is an apex institution under the MoHFW that functions as a national focal point for disease control and prevention. The NCDC provides technical support to the states and union territories for disease surveillance, outbreak investigation, laboratory diagnosis, training, research, and policy formulation. The NCDC also coordinates with various national and international agencies for HEP.
- **Disaster Management Act, 2005:** The Disaster Management Act, 2005 is a legislation that lays down the legal basis for handling disasters in India. It sets up various bodies and systems at the national, state, and district levels to plan and execute the disaster management policies and plans. It also specifies the duties, obligations, and punishments for different actors, organizations, and participants involved in disaster management.
- **Epidemic Diseases Act, (1897, 2020):** Epidemic Diseases Act, 1897 is a legislation that empowers the state and central governments to enact special rules and measures to stop and contain the outbreak of any harmful epidemic disease. The Epidemic Diseases (Amendment) Act, 2020 modified the 1897 Act to safeguard the healthcare workers from assault and intimidation and increased the central government's power to examine and isolate travellers during an epidemic situation.
- **The India COVID-19 Emergency Response and Health System Preparedness Package (ECRP):** The ECRP is a new scheme approved by the Union Cabinet in 2020 to mount an emergency response to slow and limit COVID-19 in India. The ECRP aims to develop diagnostics and COVID-dedicated treatment facilities, procure essential medical equipment and drugs, strengthen national and state health systems for prevention and preparedness for future disease outbreaks, set up laboratories and bolster surveillance activities, bio-security preparedness, pandemic research, community engagement, and risk communication.

These exemplify the endeavours of India to enhance its HEP levels. These are anticipated to yield significant benefits in India's strategy to be prepared for any health emergency in the future.

This document collates information on different policy approaches towards Health Emergency Preparedness in India. Health Emergency Preparedness is essential to prevent, detect, and respond to health emergencies. The state governments play a pivotal role in designing emergency action plans for their states and implementing the same during such situations. Understanding the diverse approaches taken by different states can help other states learn from the same.

2.

Policy Features

This section identifies a common set of policy features that the document referred to in section 3 (table of insights) cover. The features can be classified under the following four broad aspects of policy: 1) Risk Assessment and Planning, 2) Collaboration, 3) Technology Based Innovation, and 4) Capacity Building.

1) Risk Assessment and Planning

- a. **Disaster Management Plan:** A disaster management plan is an important tool for ensuring effective and coordinated response to disasters. It helps to identify organizational resources, designate roles and responsibilities, create procedures and policies, and organize activities that improve disaster readiness. The Disaster Management Act of 2005 has given the responsibility of creating National, State and District Level Disaster management plans to respective disaster management authorities. The National Disaster Management Plan (NDMP) of India is the first national plan prepared on the lines of the Sendai Framework, which is a global agreement to reduce disaster risk and losses by enhancing resilience and strengthening governance. State and District level plans are based on guidelines provided by National Disaster Management Authority. Most of the states have a State Disaster Management plan in place except for Haryana and Telangana (Ministry of Home Affairs, 2019)
- b. **Hazard Risk Vulnerability Assessment:** Hazard Risk Vulnerability Assessments (HRVAs) help to identify and prioritize the health risks and vulnerabilities of the populations, infrastructure, and resources present in each state. Most of the State Governments have conducted their HRVAs, based primarily on the Vulnerability Atlas of India, with additional narratives on remaining hazards and vulnerabilities drawn from various secondary sources. Several State Governments have conducted HRVAs on state-specific hazards and vulnerabilities on high resolution, which captures risks of disasters at district and sub-district levels. Gujarat was the first state to make an HRVA by a professional agency. It has also conducted special region-specific assessments for vulnerable regions. Himachal Pradesh has a similar HRV atlas in a digital format which uses specially designed, user-friendly software that enables the generation of hazard, vulnerability and risk maps of the State down to the block level. Assam State Disaster Management Authority has utilized the services of North Eastern Space Application Centre to develop GIS-based multi-hazard maps, vulnerability maps and risk maps for the city of Guwahati and the towns of Silchar and Dibrugarh, in high resolution of 1:10,000 scale (Ministry of Home Affairs, 2019).
- c. **Expansion in Early Warning Systems (EWS):** Many states and UTs have expanded their early warning systems to detect, monitor, and communicate potential health emergencies, such as outbreaks, epidemics, pandemics, and natural disasters in a faster way. Digital Mobile Radio (DMR) system of Andhra Pradesh and the iFLOW integrated flood warning system setup in Mumbai are some examples of the same. TNSMART (Tamil Nadu System for Multi-hazard Potential Impact Assessment, Alert, Emergency Response Planning and Tracking) System of Tamil Nadu is an innovative solution for information dissemination of all-important information in a user-friendly manner. It also has a Mobile Application linked with the web application to alert field personnel and communities to ensure timely dissemination

of early warning information to minimize loss of lives during life-threatening disasters like Tsunami (Ministry of Home Affairs, 2019).

2) Collaborations

- d. **Collaborations for Development and Relief:** Different states have collaborated with various stakeholders, such as the central government, international organizations, NGOs, private sector, and civil society, to enhance their health emergency preparedness and response. SAARC Disaster Management Centre (Interim Units) is one such important center for collaboration with SAARC countries. UNICEF has been one of the key collaborators working with various Indian states and is helping with School Education Programme in Andhra Pradesh and West Bengal and is helping the Kerala government to mainstream disaster risk reduction activities. The Bill and Melinda Gates Foundation is working with the Bihar government on various Health System Strengthening issues (Bill & Melinda Gates Foundation, 2014)

3) Technology Based Innovations

- e. **Innovative Measures:** A few states and UTs have implemented innovative measures to enhance their health emergency preparedness and response, such as using digital technologies, social media, artificial intelligence, drones, mobile apps, telemedicine, etc. Several States have made innovative use of technology, particularly social media, for risk communication, even though this has been limited mostly to communicate early warnings of disasters. Assam, Haryana, Kerala, Maharashtra, Odisha, and Tamil Nadu have reported extensive use of social media for communicating risks. A comprehensive GIS based map of Assam, by Assam State Disaster Management Authority has reported live incident reports for Storms, Floods, Fires, Landslides, and several other incidents (Assam Public Forum, 2023).

4) Capacity Building

- f. **Capacity building for Disaster Risk Management:** Capacity is defined as 'the combination of all the strengths, attributes and resources available within an organization, community or society to manage and reduce disaster risks and strengthen resilience'. This includes infrastructure, institutions, human knowledge and skills, and collective attributes such as social relationships, leadership, and management. NIDM (National Institute of Disaster Management) is India's premier institute to build capacity for disaster management. Many states have also built their own institutions for capacity building like the Gujarat Institute of Disaster Management. States also conduct regular training programmes for employees across all departments to build capacity in case of disasters. Plans for capacity building are also outlined in most of the State Disaster Management Plans (Ministry of Home Affairs, 2019). Aapda Mitra scheme was initiated by Andhra Pradesh Government to train 200 volunteers in flood-prone areas (APSDMA., 2018).

3. Table of Insights: Policy Initiatives for Health Emergency Preparedness

Note 1: Documents linked in the tables include original Acts, Policies, Schemes, Programmes and Guidelines. It is aimed at highlighting and enabling a comparison of relevant policy features covered in the documents, across the centre and the states. While all important documents have been identified, the coverage may not be exhaustive. Further, the 'Policy-at-a-Glance' is not aimed at analysing nuances of policies.

Note 2: Page numbers specified in the Table of Insights are as per the PDF pagination opened in a Google Chrome browser. It is possible to encounter differences in the actual page number for some of the documents when opened in a different web browser/application.

Data					Policy	Aspects					
						Risk Assessment and Planning			Collaborations	Tech-Based Innovation	Capacity Building
Geography	Per Capita Allocation of Funds from NDRF for 2023-24** (NDMIndia)	Per Capita Allocation of Funds from SDRF for 2023-24** (NDMIndia)	Disaster Risk Index (Out of 100) (NDMA)	Disaster Resilience Index (Out of 100) (NDMA)	References	Disaster Management Plan	Hazard Risk Vulnerability Assessment	Expansion in Early Warning Systems	Collaborations for Development and Relief	Innovative Measures	Capacity building for Disaster Risk Management
Centre					Incident Response System					Yes	
					National Disaster Management Plan	Yes					
					SAARC Disaster Management Center						Yes

Data					Policy	Aspects					
						Risk Assessment and Planning			Collaboratio ns	Tech-Based Innovation	Capacity Building
Geography	Per Capita Allocation of Funds from NDRF for 2023-24** (NDMIndia)	Per Capita Allocation of Funds from SDRF for 2023-24** (NDMIndia)	Disaster Risk Index (Out of 100) (NDMA)	Disaster Resilience Index (Out of 100) (NDMA)	References	Disaster Management Plan	Hazard Risk Vulnerability Assessment	Expansion in Early Warning Systems	Collaboration s for Development and Relief	Innovative Measures	Capacity building for Disaster Risk Management
					Vulnerability Atlas of India		Yes				
Andhra Pradesh	185.72	61.71	35.60	37	UNICEF School Safety Programme				Yes		
					Aapda Mitra			Yes		Yes	
					State Disaster Management Plan Volume 1	Yes					
					Flood Hazard Atlas of Andhra Pradesh		Yes				
Arunachal Pradesh	1413.57	158.77	19.30	22.6	State Disaster Management Plan	Yes		Yes (p. 109)	Yes (p. 152)		Yes (p. 89)
Assam	190.63	21.28	27.00	41.9	Assam State Disaster Management Plan	Yes					
					Flood Early Warning System			Yes	Yes		

Data					Policy	Aspects					
						Risk Assessment and Planning			Collaboratio ns	Tech-Based Innovation	Capacity Building
Geography	Per Capita Allocation of Funds from NDRF for 2023-24** (NDMIndia)	Per Capita Allocation of Funds from SDRF for 2023-24** (NDMIndia)	Disaster Risk Index (Out of 100) (NDMA)	Disaster Resilience Index (Out of 100) (NDMA)	References	Disaster Management Plan	Hazard Risk Vulnerability Assessment	Expansion in Early Warning Systems	Collaboratio ns for Development and Relief	Innovative Measures	Capacity building for Disaster Risk Management
					GIS Portal					Yes	
					Assam Health System Strengthening Project				Yes		Yes
Bihar	98.52	32.82	27.80	41.2	The State Disaster Management Plan	Yes		Yes (p 159)			Yes (p. 142)
Chhattisgarh	126.18	42.15	11.70	23.4	State Disaster Management Plan	Yes					Yes (p. 99)
Goa	60.95	20.32	5.20	25.6	Goa State Disaster Management Plan Part 1	Yes					
Gujarat	163.34	54.37	39.80	49.3	Gujarat State Disaster Management Plan	Yes					Yes (p. 186)
					Hazard Risk and Vulnerability Assessment		Yes				
					Gujarat Institute of Disaster Management						Yes
Himachal Pradesh	483.13	53.56	16.70	39.7	Himachal Pradesh State Disaster Management Plan	Yes		Yes (p. 85)			Yes (p. 183)
					Doctors for You					Yes	

Data					Policy	Aspects					
						Risk Assessment and Planning			Collaboratio ns	Tech-Based Innovation	Capacity Building
Geography	Per Capita Allocation of Funds from NDRF for 2023-24** <u>(NDMIndia)</u>	Per Capita Allocation of Funds from SDRF for 2023-24** <u>(NDMIndia)</u>	Disaster Risk Index (Out of 100) <u>(NDMA)</u>	Disaster Resilience Index (Out of 100) <u>(NDMA)</u>	References	Disaster Management Plan	Hazard Risk Vulnerability Assessment	Expansion in Early Warning Systems	Collaboration s for Development and Relief	Innovative Measures	Capacity building for Disaster Risk Management
					Preparation of Hazard, Vulnerability & Risk Analysis atlas & report for the state of HP		Yes				
Karnataka	103.06	34.27	24.00	32.9	Memorandum of Understanding				Yes		
					Karnataka State Disaster Management Plan	Yes					
					Kerala State Disaster Management Plan	Yes					
Kerala	77.59	25.72	22.00	41.9	UNICEF- KSDMA partnership project “Mainstreaming Disaster Risk resilience”				Yes		
					Hospital Disaster Management Guidelines						Yes

Data					Policy	Aspects					
						Risk Assessment and Planning			Collaboratio ns	Tech-Based Innovation	Capacity Building
Geography	Per Capita Allocation of Funds from NDRF for 2023-24** (NDMIndia)	Per Capita Allocation of Funds from SDRF for 2023-24** (NDMIndia)	Disaster Risk Index (Out of 100) (NDMA)	Disaster Resilience Index (Out of 100) (NDMA)	References	Disaster Management Plan	Hazard Risk Vulnerability Assessment	Expansion in Early Warning Systems	Collaboration s for Development and Relief	Innovative Measures	Capacity building for Disaster Risk Management
Madhya Pradesh	185.45	61.82	21.40	31.0	Madhya Pradesh State Disaster Management Plan	Yes					Yes (p. 64)
Maharashtra	224.84	74.95	44.40	44.3	State Disaster Management Plan	Yes					Yes (p. 45)
					Early Warning System for The City of Navi Mumbai			Yes	Yes		
					iFLOWS Mumbai			Yes			
Manipur	116.66	12.41	20.70	21.0	State Disaster Management Plan	Yes					Yes (p. 59)
Meghalaya	174.38	19.11	16.70	30.0	State Disaster Management Plan Volume 1	Yes					Yes (p. 67)
Mizoram	336.02	38.77	19.30	29.6	State Disaster Management Plan	Yes					Yes (p. 146)
Nagaland	164.80	17.91	19.20	21.2	Nagaland State Disaster Management Plan	Yes					
Odisha	305.82	101.82	10.70	41.7	State Disaster Management Plan	Yes					Yes (p. 180)
Punjab	142.14	47.38	14.70	30.6	State Disaster Management Plan	Yes					Yes (p. 105)
Rajasthan	161.33	53.71	18.80	39.1	State Disaster Management Plan	Yes					Yes (p. 119)
Sikkim	650.22	69.67	7.90	32.3	State Disaster Management Plan	Yes					Yes (p. 67)
					Multi- Hazards Risk Vulnerability Assessment		Yes				
Tamil Nadu	117.10	39.03	30.60	46.3	State Disaster Management Plan	Yes					Yes (p. 38)

Data					Policy	Aspects					
						Risk Assessment and Planning			Collaboratio ns	Tech-Based Innovation	Capacity Building
Geography	Per Capita Allocation of Funds from NDRF for 2023-24** (NDMIndia)	Per Capita Allocation of Funds from SDRF for 2023-24** (NDMIndia)	Disaster Risk Index (Out of 100) (NDMA)	Disaster Resilience Index (Out of 100) (NDMA)	References	Disaster Management Plan	Hazard Risk Vulnerability Assessment	Expansion in Early Warning Systems	Collaboration s for Development and Relief	Innovative Measures	Capacity building for Disaster Risk Management
					TN SMART		Yes	Yes	Yes	Yes	
Telangana	103.96	34.65	14.20	30.4	Telangana State Heat Wave Action Plan	Yes					Yes (p. 47)
Tripura	146.61	15.43	17.10	40.8	State Disaster Management Plan	Yes					Yes (p. 50)
Uttar Pradesh	72.37	24.10	33.10	30.3	Uttar Pradesh State Disaster Management Plan	Yes					Yes (p. 35)
Uttarakhand	710.15	79.06	22.20	36.5	Uttarakhand State Disaster Management Plan	Yes					Yes (p. 46)
					Uttarakhand Disaster Risk Assessment		Yes				
West Bengal	90.02	30.04	37.70	36.4	State Disaster Management Plan	Yes					
					Capacity Building and Training						

** Per Capita Allocation of Funds from NDRF and SDRF for 2023-24 in INR has been calculated as : total allocation of NDRF and SDRF divided by total population of that particular state. Population number has been taken from https://uidai.gov.in/images/StateWiseAge_AadhaarSat_Rep_31032023_Projected-2023-Final.pdf

4.

List of Acronyms

- HEP: Health Emergency Preparedness
- NHP: National Health Policy
- NDMP: National Disaster Management Plan
- IDSP: Integrated Disease Surveillance Programme
- NCDC: National Centre for Disease Control
- NGO: Non-Government Organisation
- SAARC: South Asian Association for Regional Cooperation
- UNICEF: United Nations International Children's Emergency Fund
- EWS: Early Warning Systems
- HRVAs: Hazard Risk Vulnerability Assessments
- DMR: Digital Mobile Radio system
- TNSMART: Tamil Nadu System for Multi-hazard Potential Impact Assessment, Alert, Emergency Response Planning and Tracking
- iFLOW: Integrated Flood Warning System Setup
- GIS: Geographical Information System
- ECRP: Emergency Response and Health System Preparedness Package

5.

1. Ministry of Home Affairs. (2019). Disaster Risks and Resilience in India. In *NDM*.
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